PROJECT REPORT

CS555

Foundations of Machine Learning

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Topic: Understanding Engagement Level

Dataset used: Emotional Monitoring Dataset with Target from Kaggle

1. Research Scenario and Questions:

Research Scenario: This research aims to evaluate how some physiological and environmental factors influence the engagement level of an individual. This level of engagement is critical for everything from workplaces to classrooms to labs; and it powers productivity, attention, and performance. By looking at how factors like heart rate, skin conductance, and ambient noise level influence engagement, we can gain valuable insights into improving engagement in these settings. Finally, we were interested in how emotional and cognitive states (e.g., whether the person is focused or distracted) can determine engagement over time.

Research Questions:

- *Do emotional and cognitive states impact engagement levels, and if so, how?*
- *What factors, such as heart rate and environmental conditions, are the most significant predictors of a person's engagement level?*
- *Does engagement level change with physiological responses like heart rate or skin conductance?*
- *How does the surrounding environment (ambient noise or lighting) influence a person's engagement?*
- *Do emotional and cognitive states impact engagement levels, and if so, how?*

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2. Description of the Dataset

Dataset : The dataset contains measurements from individuals that describe their physiological responses, environmental conditions, and engagement levels. These measurements are taken across various scenarios where the individual's engagement is of interest.

Variables:

- **HeartRate**: This represents the number of heartbeats per minute. A higher heart rate can indicate stress or high engagement.
- **SkinConductance**: This measures the conductivity of the skin, which changes with sweat levels—often an indicator of emotional or physiological responses to stress.
- **EEG**: This provides data on brain activity, which can give insights into cognitive engagement.
- **Temperature**: Body temperature, which may vary depending on stress or emotional states.
- **PupilDiameter**: The size of the pupils, which can reflect focus or attention levels.
- **SmileIntensity**: The intensity of a person's smile, potentially indicating emotional states like happiness or engagement.
- **FrownIntensity**: Similarly, the intensity of frowning could indicate stress or frustration.
- **CortisolLevel**: Cortisol is a hormone linked to stress, and its level might correlate with engagement or emotional state.
- **ActivityLevel**: How physically active the person is, which could be related to engagement.
- **AmbientNoiseLevel**: The level of noise in the surrounding environment, which could affect engagement.
- **LightingLevel**: The amount of light in the environment, which could influence how well an individual focuses.
- **EmotionalState**: A categorical measure of the person's emotional state (engaged, partially engaged, or disengaged).
- **CognitiveState**: Whether the person is focused or distracted.
- **EngagementLevel**: This is the dependent variable, measuring how engaged the person is, which is our primary interest.

Quick view of the data:

	HeartRate	SkinConductance	EEG	Temperature	PupilDiameter	SmileIntensity	FrownIntensity	CortisolLevel	ActivityLevel	AmbientNoiseLevel	LightingLevel	EmotionalState	CognitiveState	EngagementLevel
1	61	8.93720407	11.7949463	36.50172	3.330181	0.689237525	0.1890235058	0.6030351	136	59	394	engaged	distracted	3
2	60	12.63539735	19.1514124	36.61891	3.428995	0.561056311	0.0913670861	0.5666712	155	39	479	engaged	distracted	1
3	81	3.66002832	6.2260979	36.17690	2.819286	0.417951084	0.2273549576	1.4224747	55	30	832	partially engaged	focused	3
4	119	0.56307022	4.5429679	37.20529	2.192961	0.140185794	0.5029649863	1.6690448	39	40	602	disengaged	focused	3
5	118	0.47737776	0.9962094	37.24812	2.450139	0.064471019	0.6956043846	1.8540758	10	42	908	disengaged	focused	3
6	77	9.02455178	19.6701599	36.80567	3.728565	0.672491423	0.1339898362	0.6288988	151	47	449	engaged	focused	3
7	83	4.04982983	6.3471925	36.29936	2.721967	0.419392057	0.2581759464	1.3577255	58	42	504	partially engaged	focused	3
8	112	1.32230854	3.4779259	37.35205	2.341349	0.078561543	0.6184561023	1.6274088	9	55	388	disengaged	focused	3
9	92	2.20533876	7.2375014	36.08201	2.673052	0.463995962	0.2768094766	1.3049355	85	34	598	partially engaged	focused	2
10	89	4.76132311	8.2181640	36.13548	2.561627	0.329945113	0.3565325854	1.2940550	77	59	663	partially engaged	distracted	3
11	113	1.71151601	4.3738126	37.29063	2.262272	0.163289608	0.7248005539	1.7001626	6	59	381	disengaged	focused	3
12	65	13.15621771	11.7920985	36.54110	3.328358	0.712188384	0.1085318564	0.6862118	150	35	648	engaged	focused	3
13	79	8.82255915	16.4757849	36.98314	3.510375	0.993438423	0.0604498790	0.5154341	138	55	906	engaged	focused	3
14	76	14.53603368	16.5048335	36.83656	3.178929	0.746304839	0.0734165494	0.7354784	119	50	985	engaged	focused	1
15	92	4.71662498	7.7240794	36.46225	2.786177	0.313927263	0.3105738787	1.2878695	85	47	474	partially engaged	focused	2
16	62	11.22876149	12.5961410	36.79087	3.240952	0.723222636	0.0245411399	0.5186503	142	33	370	engaged	focused	3

Figure 2.1

Data Cleaning: The dataset has been cleaned by:

- Ensuring that all variables are in the correct format (e.g., numerical for physiological measures, categorical for emotional and cognitive states).
- Checking for missing data and filling or removing values where necessary.
- Converting categorical variables like EmotionalState and CognitiveState into factors for analysis.

Dataset Link: https://www.kaggle.com/datasets/ziya07/emotional-monitoring-dataset

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3. Statistical Methods

Statistical methods used to investigate the research question:

- **Hypothesis Testing:** Evaluates the validity of a claim or hypothesis based onsample data.
- **Correlation Tests:** Measures the strength and direction of the linear relationship between two variables.
- **Simple Linear Regression:** Models the relationship between a dependentvariable and a single independent variable.
- **Multiple Linear Regression:** Models the relationship between a dependentvariable and multiple independent variables.(model the relationship between EngagementLevel and several predictors such as HeartRate, SkinConductance, AmbientNoiseLevel, and LightingLevel. The goal is to see how these factors together influence engagement.)
- **ANOVA (Analysis of Variance)**: Compares means among more than two groupsto detect significant differences.(Since EmotionalState and CognitiveState are categorical variables, ANOVA will help us determine if there are substantial differences in engagement levels

between different groups (e.g., "focused" vs. "distracted").)

- **Correlation Analysis**: We will calculate the correlation between physiological and environmental variables and EngagementLevel to understand the strength and direction of relationships.
- **Visualization**: We will use scatter plots, boxplots, and heatmaps to visually explore the relationships between variables and help identify patterns or trends in the data.

4. *Results*

a. Data

We have the columns:

HeartRate, SkinConductance, EEG, Temperature, PupilDiameter, SmileIntensity, FrownIntensity, CortisolLevel, ActivityLevel, AmbientNoiseLevel, LightingLevel, EmotionalState, CognitiveState, EngagementLevel.

```
'data.frame':   1000 obs. of  14 variables:
 $ HeartRate        : int  61 60 81 119 118 77 83 112 92 89 ...
 $ SkinConductance  : num  8.937 12.635 3.66 0.563 0.477 ...
 $ EEG              : num  11.795 19.151 6.226 4.543 0.996 ...
 $ Temperature      : num  36.5 36.6 36.2 37.2 37.2 ...
 $ PupilDiameter    : num  3.33 3.43 2.82 2.19 2.45 ...
 $ SmileIntensity   : num  0.6892 0.5611 0.418 0.1402 0.0645 ...
 $ FrownIntensity   : num  0.189 0.0914 0.2274 0.503 0.6956 ...
 $ CortisolLevel    : num  0.603 0.567 1.422 1.669 1.854 ...
 $ ActivityLevel    : int  136 155 55 39 10 151 58 9 85 77 ...
 $ AmbientNoiseLevel: int  59 39 30 40 42 47 42 55 34 59 ...
 $ LightingLevel    : int  394 479 832 602 908 449 504 388 598 663 ...
 $ EmotionalState   : chr  "engaged" "engaged" "partially engaged" "disengaged" ...
 $ CognitiveState   : chr  "distracted" "distracted" "focused" "focused" ...
 $ EngagementLevel  : int  3 1 3 3 3 3 3 3 2 3 ...
```

b. Exploratory Data Analysis (EDA)

Exploratory Data Analysis and Visualization the Distribution of Engagement Level

These graphs provide insight into how engagement levels vary across cognitive and emotional states, indicating patterns that might help in understanding behavior or performance metrics in different conditions.

Distribution of Engagement Level:

- This histogram shows the count of instances for different levels of "Engagement Level."
- The graph suggests that Engagement Level 3 has the highest count, while Engagement Level 1 has the lowest count. Engagement Level 2 lies in between.

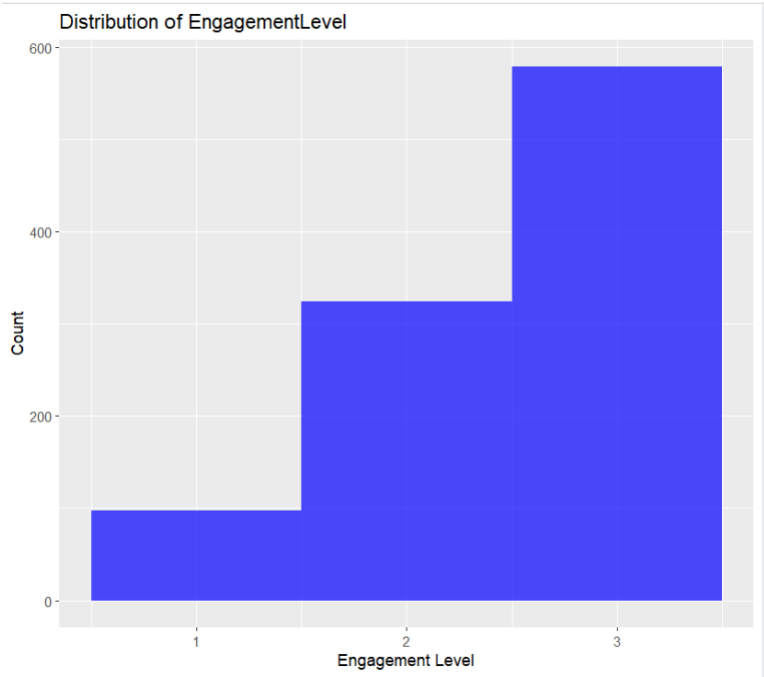
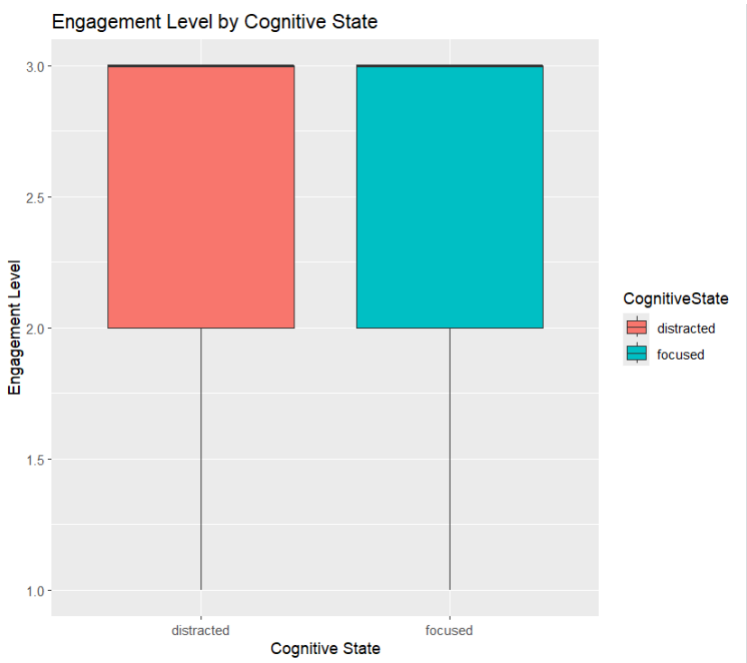


Figure: Distribution of Engagement level

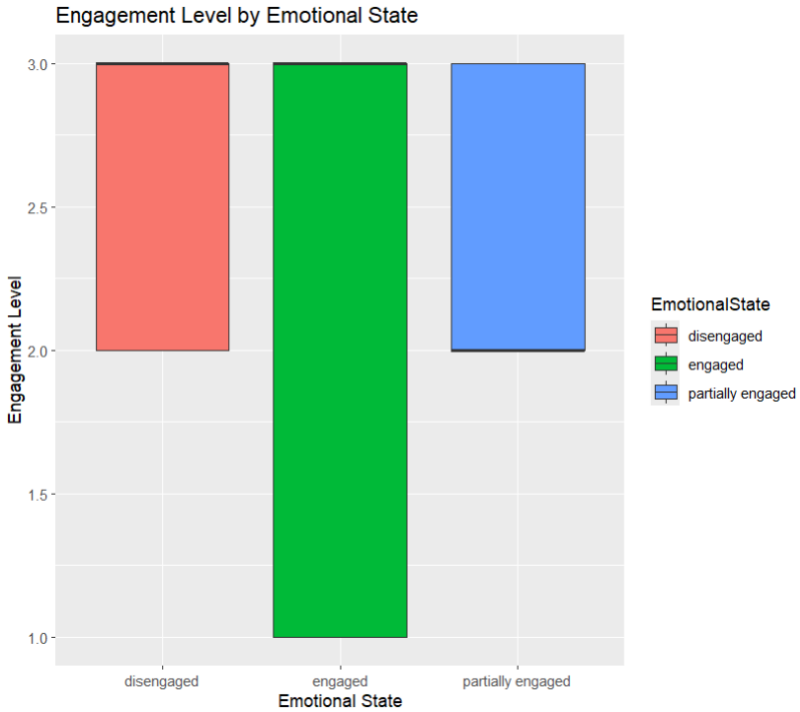
Engagement Level by Cognitive State :

- This bar plot compares Engagement Levels between two cognitive states: "distracted" and "focused."
- The levels seem to be comparable across these states, with no significant difference in Engagement Level between individuals who are distracted and those who are focused.



Engagement Level by Emotional State :

- This bar plot categorizes Engagement Level based on Emotional State: "disengaged," "engaged," and "partially engaged."
- Engagement Level appears to be highest for the "engaged" emotional state, followed by "partially engaged," and lowest for "disengaged."



C. Hypothesis Testing and ANOVA

- **Answering Research question: "***Do emotional and cognitive states impact engagement levels, and if so, how?"*

Test for Differences in EngagementLevel by CognitiveState (ANOVA)

Hypotheses:

H0: Mean engagement levels across CognitiveState groups are equal.

$H1$: Mean engagement levels differ across CognitiveState groups.

```
> #HYPOTHESIS
> #Test for Differences in EngagementLevel by CognitiveState (ANOVA)
> #Hypotheses:
> #H0: Mean engagement levels across CognitiveState groups are equal.
> #H1: Mean engagement levels differ across CognitiveState groups.
> # Perform ANOVA
> anova_result <- aov(EngagementLevel ~ CognitiveState, data = data)
> summary(anova_result)
                Df Sum Sq Mean Sq F value Pr(>F)
CognitiveState   1    0.2  0.2255   0.507  0.476
Residuals      998  443.5  0.4443
> # Extract the p-value for the CognitiveState variable
> anova_summary <- summary(anova_result)
> p_value <- anova_summary[[1]][["Pr(>F)"]][1]
> # Check if the p-value is significant
> if (p_value < 0.05) {
+   print("Significant result, performing Tukey HSD post-hoc test:")
+   print(TukeyHSD(anova_result))
+ } else {
+   print("No significant differences found.")
+ }
[1] "No significant differences found."
```

The null hypothesis (H_0) assumes that the mean engagement levels across CognitiveState groups ("distracted" and "focused") are equal.

The p-value for the test is 0.476, which is greater than the common significance level (e.g., 0.05).

Since the p-value is not significant, we fail to reject the null hypothesis (H_0).

This implies there is no evidence of a statistically significant difference in mean engagement levels between the two CognitiveState groups ("distracted" and "focused").

The test result implies that the CognitiveState (distracted or focused) does not significantly affect the EngagementLevel in your data. This might suggest that engagement level is similar regardless of whether someone is distracted or focused.

Similarly with Emotional State:

```
> # ANOVA for Engagement Level by Emotional State
> anova_emotional <- aov(EngagementLevel ~ EmotionalState, data = data)
> summary(anova_emotional)
                Df Sum Sq Mean Sq F value Pr(>F)
EmotionalState   2    2.4  1.1908    2.69 0.0683 .
Residuals      997  441.3  0.4426
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> # Checking significance
> p_value_emotional <- summary(anova_emotional)[[1]][["Pr(>F)"]][1]
> if (p_value_emotional < 0.05) {
+   print("Significant result for Emotional State! Performing Tukey HSD post-hoc test:")
+   print(TukeyHSD(anova_emotional))
+ } else {
+   print("No significant differences found for Emotional State.")
+ }
[1] "No significant differences found for Emotional State."
```

d. Simple Linear Regression

```
Simple Linear Regression for Predictor: HeartRate

Call:
lm(formula = formula, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.4670 -0.4958  0.4813  0.5315  0.5603 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 2.353467   0.110744  21.251   <2e-16 ***
HeartRate   0.001437   0.001216   1.182    0.237    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6663 on 998 degrees of freedom
Multiple R-squared:  0.001398,	Adjusted R-squared:  0.0003979 
F-statistic: 1.398 on 1 and 998 DF,  p-value: 0.2374
```

Predictor: HeartRate

- Intercept: 2.35
- Slope: 0.0014
- R-squared: 0.0014 (0.14%)
- P-value: 0.237 (Not significant)
- Conclusion: HeartRate is not a significant predictor of EngagementLevel.

```
Simple Linear Regression for Predictor: SkinConductance

Call:
lm(formula = formula, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.2948 -0.5867  0.3775  0.5152  0.8879 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      2.660948   0.031543  84.358  < 2e-16 ***
SkinConductance -0.036617   0.004902  -7.469 1.76e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6489 on 998 degrees of freedom
Multiple R-squared:  0.05294,	Adjusted R-squared:  0.052 
F-statistic: 55.79 on 1 and 998 DF,  p-value: 1.755e-13


Simple Linear Regression for Predictor: AmbientNoiseLevel

Call:
lm(formula = formula, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.5146 -0.4829  0.4922  0.5239  0.5510 

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)       2.381104   0.110434  21.561   <2e-16 ***
AmbientNoiseLevel 0.002263   0.002431   0.931    0.352    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6665 on 998 degrees of freedom
Multiple R-squared:  0.0008672, Adjusted R-squared:  -0.0001339 
F-statistic: 0.8663 on 1 and 998 DF,  p-value: 0.3522
```

Predictor: SkinConductance

- Intercept: 2.66
- Slope: -0.0366
- R-squared: 0.053 (5.3%)
- P-value: 1.76e-13 (Highly significant)
- Conclusion: SkinConductance has a statistically significant negative relationship with EngagementLevel.

Predictor: AmbientNoiseLevel

- Intercept: 2.38
- Slope: 0.0023
- R-squared: 0.0009 (0.09%)
- P-value: 0.352 (Not significant)
- Conclusion: AmbientNoiseLevel is not a significant predictor of EngagementLevel.

```
Simple Linear Regression for Predictor: LightingLevel

Call:
lm(formula = formula, data = data)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.5390 -0.4810  0.4728  0.5244  0.5790 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)   2.3685888  0.0729612  32.464   <2e-16 ***
LightingLevel 0.0001730  0.0001065   1.623    0.105    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6659 on 998 degrees of freedom
Multiple R-squared:  0.002634,	Adjusted R-squared:  0.001635 
F-statistic: 2.636 on 1 and 998 DF,  p-value: 0.1048
```

Predictor: LightingLevel

- Intercept: 2.37
- Slope: 0.00017
- R-squared: 0.0026 (0.26%)
- P-value: 0.105 (Not significant)
- Conclusion: LightingLevel is not a significant predictor of EngagementLevel.

From the outputs we derive that:

Among the predictors, only SkinConductance shows a significant effect on EngagementLevel, explaining 5.3% of the variance.

Other predictors (HeartRate, AmbientNoiseLevel, LightingLevel) do not significantly impact EngagementLevel, as evidenced by their high p-values and very low R-squared values.

e. **Multiple Linear Regression**

```
Call:
lm(formula = EngagementLevel ~ HeartRate + SkinConductance +
    AmbientNoiseLevel + LightingLevel, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-1.4688 -0.5608  0.2351  0.5362  1.2701

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)        4.3206684  0.2464923  17.529   <2e-16 ***
HeartRate         -0.0169809  0.0019664  -8.635   <2e-16 ***
SkinConductance   -0.0936976  0.0081477 -11.500   <2e-16 ***
AmbientNoiseLevel  0.0008896  0.0022865   0.389    0.697
LightingLevel      0.0001497  0.0001002   1.494    0.135
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.6259 on 995 degrees of freedom
Multiple R-squared:  0.1216,    Adjusted R-squared:  0.118
F-statistic: 34.43 on 4 and 995 DF,  p-value: < 2.2e-16
```

Model Summary:

- R-squared: 0.1216 (12.16%)
- Adjusted R-squared: 0.118 (11.8%)
- Residual Standard Error: 0.6259
- F-statistic: 34.43 ($p < 2.2e-16$)

Coefficients:

1. Intercept:

   Estimate: 4.32

   Interpretation: Baseline EngagementLevel when all predictors are zero.

2. HeartRate:

   Estimate: -0.017

   P-value: < 2e-16 (Highly significant)

   Interpretation: A unit increase in HeartRate is associated with a 0.017 decrease in EngagementLevel.

3. SkinConductance:

   Estimate: -0.094

   P-value: < 2e-16 (Highly significant)

   Interpretation: A unit increase in SkinConductance is associated with a 0.094 decrease in EngagementLevel.

4.

5. AmbientNoiseLevel:

   Estimate: 0.0009

   P-value: 0.697 (Not significant)

   Interpretation: AmbientNoiseLevel does not significantly affect EngagementLevel.

6. LightingLevel:

   Estimate: 0.00015

   P-value: 0.135 (Not significant)

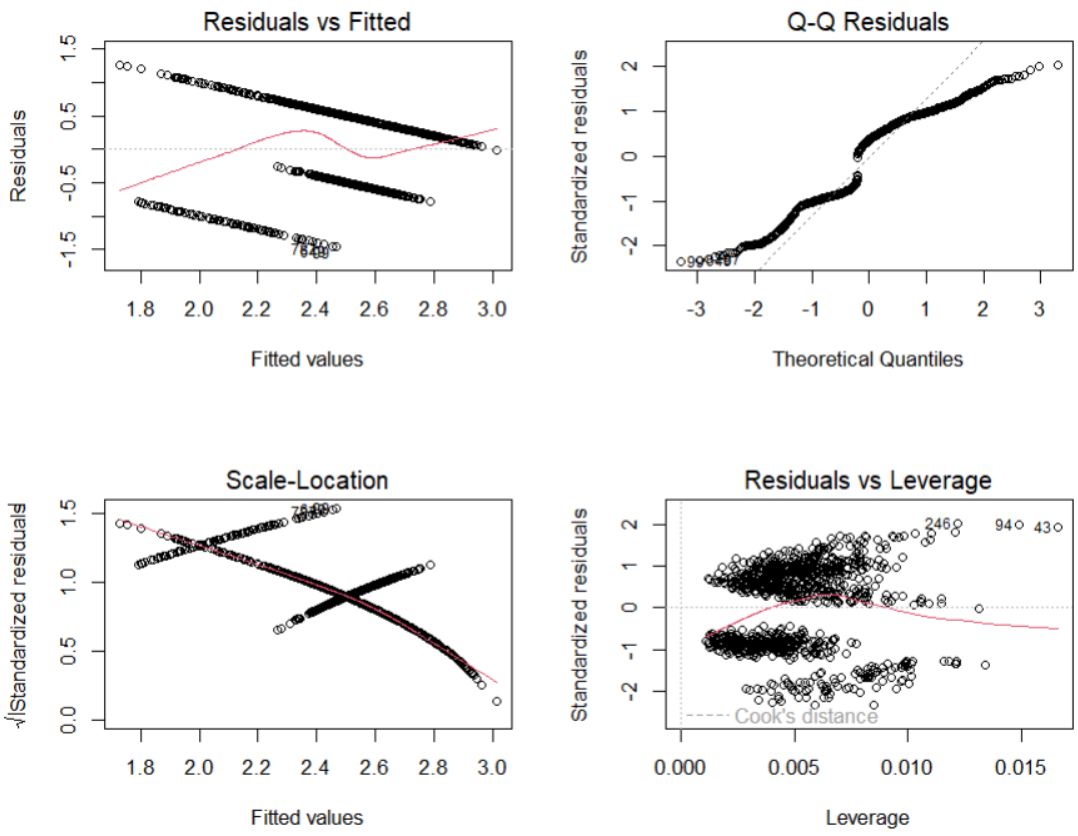
   Interpretation: LightingLevel does not significantly affect EngagementLevel.

The model explains 12.16% of the variance in EngagementLevel, indicating some predictive power but leaving significant variance unexplained.

HeartRate and SkinConductance are significant predictors, both negatively associated with EngagementLevel.

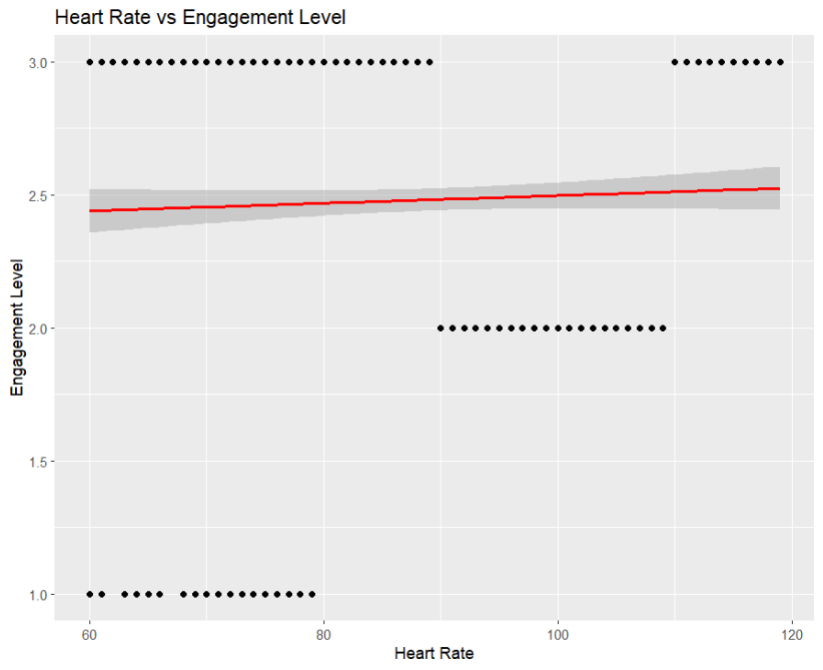
AmbientNoiseLevel and LightingLevel are not significant predictors in this model.

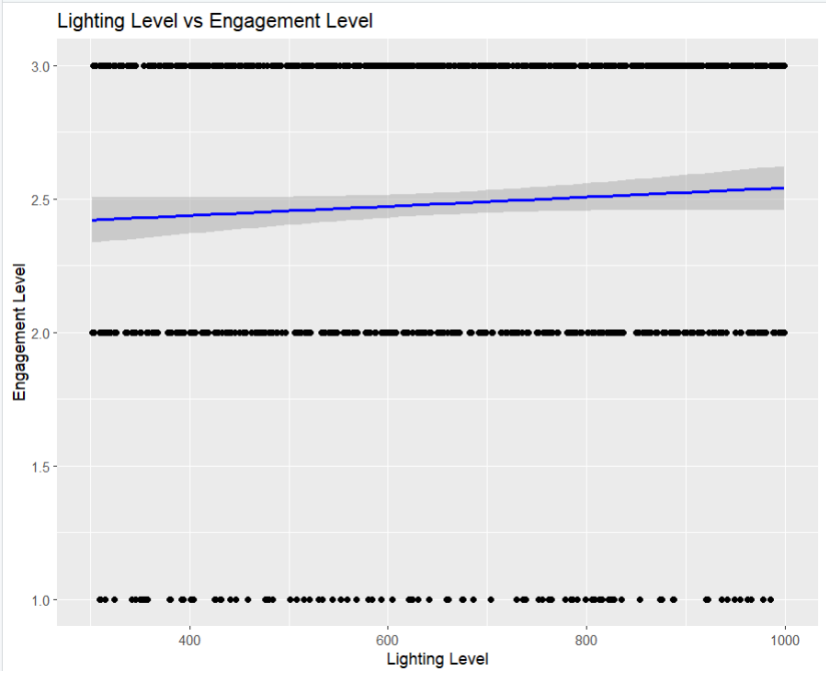
f. Diagnostic Plots for Model Validation



- Residuals vs Fitted: Non-random patterns suggest the model might not adequately capture the data's variability, potentially indicating a non-linear relationship or omitted variables.
- Q-Q Plot: Deviation from the theoretical quantiles suggests that residuals are not normally distributed.
- Scale-Location Plot: The trend in the spread of residuals indicates heteroscedasticity, violating an assumption of linear regression.
- Residuals vs Leverage: Some high-leverage points (e.g., 940 and 430) may disproportionately influence the model.

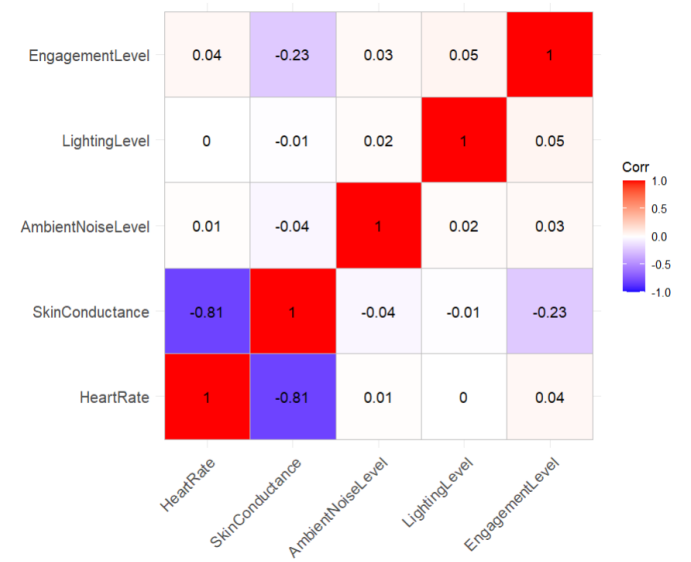
g. **Regression Analysis**



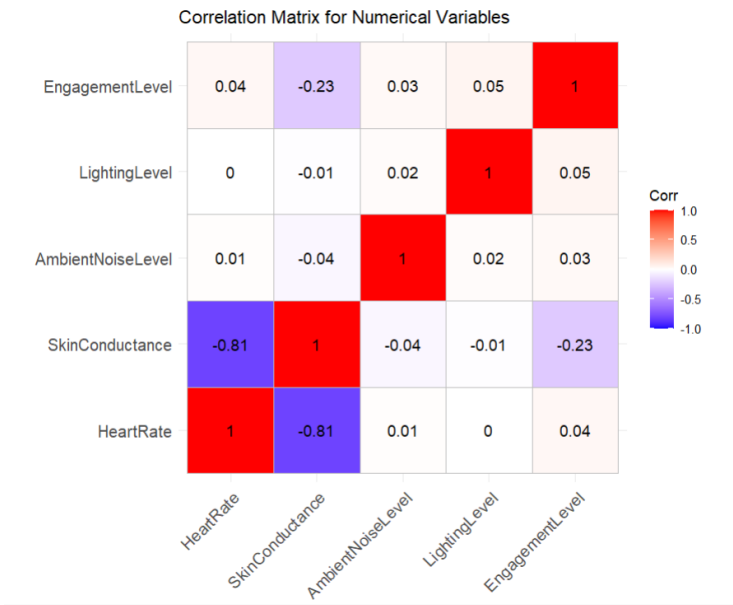


- **Heart Rate vs Engagement Level**: The red regression line indicates a slight positive relationship between heart rate and engagement level. However, the spread of data points suggests a weak correlation.
- **Lighting Level vs Engagement Level**: The blue regression line also shows a slight positive trend between lighting level and engagement level. Like the first graph, the weak slope indicates minimal influence.

h. Correlation Matrix




```
[1] "Correlation Matrix:"
> print(cor_matrix)
                   HeartRate SkinConductance AmbientNoiseLevel LightingLevel
HeartRate          1.000000000     -0.81381878        0.01185298  -0.001663746
SkinConductance   -0.813818782      1.00000000       -0.03769434  -0.010089673
AmbientNoiseLevel  0.011852983     -0.03769434        1.00000000   0.020632962
LightingLevel     -0.001663746     -0.01008967        0.02063296   1.000000000
EngagementLevel    0.037396419     -0.23009608        0.02944910   0.051322814
                  EngagementLevel
HeartRate              0.03739642
SkinConductance       -0.23009608
AmbientNoiseLevel      0.02944910
LightingLevel          0.05132281
EngagementLevel        1.00000000
```



- **Engagement Level**: Weak correlations with other variables (e.g., 0.04 with heart rate, -0.23 with skin conductance).
- **Heart Rate and Skin Conductance**: A strong negative correlation (-0.81) suggests that an increase in one is associated with a decrease in the other.
- **Lighting and Engagement Level**: Near-zero correlation indicates lighting has negligible direct influence on engagement.

Conclusion:

1. Exploratory Data Analysis (EDA)

Distribution of Engagement Level:
The histogram of engagement levels shows that Engagement Level 3 has the highest count, followed by Engagement Level 2, and Engagement Level 1 has the lowest count.

Engagement Level by Cognitive State:
The bar plot shows comparable engagement levels for the cognitive states "distracted" and "focused." No significant difference is observed between these states in terms of engagement levels.

Engagement Level by Emotional State:
The bar plot indicates that individuals categorized as "engaged" have the highest engagement levels, followed by "partially engaged," and the lowest engagement levels are observed for the "disengaged" state.

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2. Hypothesis Testing and ANOVA

Test for Differences in Engagement Level by Cognitive and Emotional State:

- Null Hypothesis (H_0): The mean engagement levels across cognitive states are equal, **P-value = 0.476**
- Since the p-value is greater than 0.05, we fail to reject the null hypothesis.
- **Conclusion:** There is no statistically significant difference in engagement levels between "distracted" and "focused" cognitive states.

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3. Simple Linear Regression

Summary of Predictor Variables:

Predictor	Intercept	Slope	R-squared	P-value	Conclusion
HeartRate	2.35	0.0014	0.0014	0.237 (NS)	Not a significant predictor
SkinConductance	2.66	-0.0366	0.053	1.76e-13 (***S)	Significant negative relationship
AmbientNoiseLevel	2.38	0.0023	0.0009	0.352 (NS)	Not a significant predictor
LightingLevel	2.37	0.00017	0.0026	0.105 (NS)	Not a significant predictor

- **Thus,** SkinConductance is the only significant predictor of Engagement Level, explaining 5.3% of the variance. Other predictors (HeartRate, AmbientNoiseLevel, and LightingLevel) are not significant.

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4. Multiple Linear Regression

Model Summary:

- R-squared: 0.1216 (12.16%)
- Adjusted R-squared: 0.118 (11.8%)
- F-statistic: 34.43 (**p < 2.2e-16**)
- Residual Standard Error: 0.6259

Coefficients:

Predictor	Estimate	P-value	Conclusion
Intercept	4.32	-	Baseline Engagement Level
HeartRate	-0.017	< 2e-16 (***S)	Significant negative relationship
SkinConductance	-0.094	< 2e-16 (***S)	Significant negative relationship
AmbientNoiseLevel	0.0009	0.697 (NS)	Not a significant predictor
LightingLevel	0.00015	0.135 (NS)	Not a significant predictor

Thus, HeartRate and SkinConductance are significant predictors of Engagement Level, both showing negative relationships. AmbientNoiseLevel and LightingLevel are not significant.

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5. Diagnostic Plots

- **Residuals vs Fitted Plot:** Shows non-random patterns, indicating potential issues such as omitted variables or a non-linear relationship.
- **Q-Q Plot:** Residuals deviate from the theoretical quantiles, suggesting non-normality.
- **Scale-Location Plot:** Heteroscedasticity is present, violating linear regression assumptions.
- **Residuals vs Leverage Plot:** Identifies high-leverage points (e.g., observations 940 and 430) that may influence the model disproportionately.

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6. Correlation Analysis

Correlation Matrix:

Variable	**Engagement Level**	**HeartRate**	**SkinConductance**	**LightingLevel**
Engagement Level	1.00	0.04	-0.23	0.00
HeartRate	0.04	1.00	-0.81	0.05
SkinConductance	-0.23	-0.81	1.00	-0.01
LightingLevel	0.00	0.05	-0.01	1.00

Weak correlation between Engagement Level and other variables.

Strong negative correlation between HeartRate and SkinConductance (-0.81).

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7. Visualizations

- **Heart Rate vs. Engagement Level:** Shows a slight positive relationship, but the spread of data indicates a weak correlation.
- **Lighting Level vs. Engagement Level:** The regression line shows minimal influence of lighting on engagement.

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8. Summary of Findings

- SkinConductance is the most significant predictor of Engagement Level, with a negative relationship.
- HeartRate is a significant predictor in the multiple regression model but not in simple regression.
- Environmental factors (AmbientNoiseLevel and LightingLevel) have negligible impacts on Engagement Level.
- Emotional state significantly influences engagement, while cognitive state does not.
- Diagnostic plots reveal issues with heteroscedasticity, normality, and high-leverage points, suggesting room for model improvement.

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9. Limitations:

1. *Limited Dataset Context:*
   The dataset lacks detailed contextual information, such as the specific scenarios in which measurements were taken or individual differences (e.g., age, prior stress levels, personality traits) that might influence engagement.
2. *Potential Multicollinearity:*
   Strong correlations between some variables (e.g., heart rate and skin conductance with a correlation of -0.81) could indicate multicollinearity, which may bias the regression model coefficients and interpretations.
3. *Sample Size and Generalizability:*
   If the sample size is small or the data comes from a homogeneous group, the results might not generalize to broader populations or different environments.
4. *Simplistic Assumptions:*
   The analysis assumes linear relationships between predictors and engagement levels. However, engagement is likely influenced by more complex, non-linear dynamics that were not captured in the study.
5. *Potential Noise in Measurement:*
   Variables (smile intensity, pupil diameter,ambient noise) may have variability due to external factors, reducing the reliability of the analysis.